

2 (a)	resistance = potential difference / current	B1	[1]
(b) (i)	metal wire in series with power supply and ammeter voltmeter in parallel with metal wire rheostat in series with power supply or potential divider arrangement or variable power supply	B1 B1 B1	[3]
(ii) 1.	intercept on graph	B1	[1]
2.	scatter of readings about the best fit line	B1	[1]
(iii)	correction for zero error explained use of V and corrected I values from graph resistance = $V/I = 22.(2)\Omega$ [e.g. $4.0 / 0.18$]	B1 C1 A1	[3]
(c)	$R = 6.8 / 0.64 = 10.625$	C1	
	$\%R = \%V + \%I$ $= (0.1 / 6.8) \times 100 + (0.01 / 0.64) \times 100$ $= 1.47\% + 1.56\%$	C1	
	$\Delta R = 0.0303 \times 10.625 = 0.32\Omega$ $R = 10.6 \pm 0.3\Omega$	A1	[3]

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2 (a) pressure = force / area

P1 [1]